

CM0246 Discrete Structures

Representing Relations

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Preliminaries

Textbook

Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Matrices

Definition

A **matrix** is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ matrix.

$$\mathbf{A} = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Matrices

Definition

Let $\mathbf{A}_{m \times k}$ and $\mathbf{B}_{k \times n}$ be two matrices. The product $\mathbf{AB}$ is the matrix $m \times n$ defined by

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & & \vdots \\ \mathbf{a_{i1}} & \mathbf{a_{i2}} & \cdots & \mathbf{a_{ik}} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mk} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & \mathbf{b_{1j}} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & \mathbf{b_{2j}} & \cdots & b_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{k1} & b_{k2} & \cdots & \mathbf{b_{kj}} & \cdots & b_{kn} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \mathbf{c_{ij}} & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{bmatrix}$$

where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj}.$$

Matrices

Definition

Let $\mathbf{A} = [a_{ij}]$ be a $m \times n$ matrix. The **transpose** of $\mathbf{A}$, denoted by $\mathbf{A}^t$, is the $n \times m$ matrix obtained by interchanging the rows and columns of $\mathbf{A}$.

Matrices

Boolean operations

$$b_1 \wedge b_2 = \begin{cases} 1, & \text{if } b_1 = b_2 = 1; \\ 0, & \text{otherwise,} \end{cases}$$

$$b_1 \vee b_2 = \begin{cases} 1, & \text{if } b_1 = 1 \text{ or } b_2 = 1; \\ 0, & \text{otherwise.} \end{cases}$$

Matrices

Definition

Let $\mathbf{A}$ and $\mathbf{B}$ be $m \times n$ Booleans matrices.

- (a) The **join** (Spanish: **unión**) of $\mathbf{A}$ and $\mathbf{B}$, denoted $\mathbf{A} \vee \mathbf{B}$, is the Boolean matrix with (i, j) -th entry $a_{ij} \vee b_{ij}$.
- (b) The **meet** (Spanish: **intersección**) of $\mathbf{A}$ and $\mathbf{B}$, denoted $\mathbf{A} \wedge \mathbf{B}$, is the Boolean matrix with (i, j) -th entry $a_{ij} \wedge b_{ij}$.

Matrices

Example

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix},$$

Matrices

Example

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix},$$

$$\mathbf{A} \vee \mathbf{B} = \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$

Matrices

Example

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix},$$

$$\mathbf{A} \vee \mathbf{B} = \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$

$$\mathbf{A} \wedge \mathbf{B} = \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Matrices

Definition

Let $A_{m \times k}$ and $B_{k \times n}$ be two Boolean matrices. The **Boolean product** of A and B , denoted $A \odot B$, is the Boolean matrix $m \times n$ with (i, j) -th entry

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \cdots \vee (a_{ik} \wedge b_{kj}).$$

Matrices

Example (product of Boolean matrices)

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix},$$

Matrices

Example (product of Boolean matrices)

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix},$$

$$\begin{aligned} \mathbf{A} \odot \mathbf{B} &= \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix} \\ &= \begin{bmatrix} 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \\ 0 \vee 0 & 0 \vee 1 & 0 \vee 1 \\ 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}. \end{aligned}$$

Representing Relations Using Boolean Matrices

Representing of relations using Boolean matrices

Let R be a relation from a A to B . The relation R can be represented by the Boolean matrix $\mathbf{M}_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1, & \text{if } (a_i, b_j) \in R; \\ 0, & \text{otherwise.} \end{cases}$$

Representing Relations Using Boolean Matrices

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Let R be a relation from a A to B . The relation R can be represented by the **Boolean** matrix $\mathbf{M}_R = [m_{ij}]$, where

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Example

Whiteboard.

Representing Relations Using Boolean Matrices

Properties of relations from their Boolean matrix representation

See slides § 8.3 for the 6th ed. of Rosen's textbook.

Representing Relations Using Boolean Matrices

Boolean matrix representing of the combination of relations

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2},$$

$$\mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2},$$

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S.$$

Representing Relations Using Boolean Matrices

Example

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \end{matrix}, \quad M_S = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix} \end{matrix},$$

$$M_{S \circ R} = M_R \odot M_S = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \end{matrix}.$$

Representing Relations Using Digraphs

Definition

A **digraph** (or **directed graph**) consists of a set V of vertices together with a set E of ordered pairs of elements of V called edges.

Representing Relations Using Digraphs

Definition

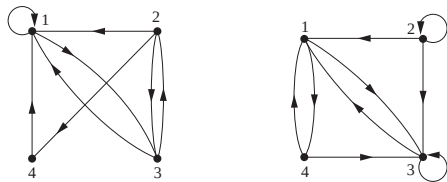
A **digraph** (or **directed graph**) consists of a set V of vertices together with a set E of ordered pairs of elements of V called edges.

Representing relations using digraphs

The relation R on a set A is represented by a digraph where $V = A$ and (a, b) is an edge if $(a, b) \in R$.

Representing Relations Using Digraphs[†]

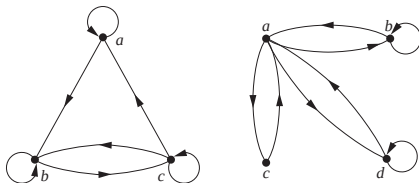
Example



[†]Figure source: (Rosen [1988] 2012, § 9.3, Figs. 4 and 5).

Representing Relations Using Digraphs[†]

Properties of relations



[†]Figure source: (Rosen [1988] 2012, § 9.3, Fig. 6).

References

- Kenneth H. Rosen [1988] (2004). *Matemática Discreta y sus Aplicaciones*. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).
- [1988] (2012). *Discrete Mathematics and Its Applications*. 7th ed. McGraw-Hill (cit. on pp. 21, 22).